

Confidence Analysis from Speech Signals

A Machine Learning Approach Using Acoustic Features

S. Udhaya Sankari*, Sai Rithvik Ambati & Shaman K Vidymananda
 Department of Computer Science & Engineering,
 Amrita School of Computing, Bengaluru
 Amrita Vishwa Vidyapeetham, India

*Corresponding Author: bl.en.u4cse23150@bl.students.amrita.edu

Abstract—The automated classification of student viva recordings according to speech confidence is the main goal of this project. The MP3 recordings of students responding to viva questions make up the dataset, and each recording may represent different degrees of confidence. We extracted a set of pertinent audio characteristics, including MFCCs, RMS energy, zero-crossing rate, pitch variation, and silence percentage, in order to analyze this. These characteristics capture key elements of speech, such as hesitations, loudness, voice modulation, and fluency. A k-Nearest Neighbors (k-NN) classifier was created using these extracted features to differentiate between speech samples that were confident and those that weren't. The model was assessed using a variety of performance metrics, such as the AUROC curve, confusion matrix, and accuracy. Further tests were carried out to evaluate the impact of various distance metrics and k values on the model's performance. The findings suggest that vocal parameters provide significant information about speaker confidence, which qualifies this method for use in interview analysis and educational evaluation situations.

Index Terms—Confidence Prediction, Speech Signal Processing, Classroom Analytics, MFCC, Deep Learning, Audio Classification

I. INTRODUCTION

Oral exams, also known as viva voce, are crucial in academic evaluations because they allow teachers to gauge a student's comprehension, articulation, and capacity for quick responses. Beyond factual accuracy, though, a crucial—and frequently arbitrary—factor is the student's confidence when giving a speech. Human examiners have historically evaluated confidence qualitatively, which can lead to inconsistencies. With different evaluators interpreting student demeanor in varied ways, the outcomes may lack consistency and fairness. This becomes particularly problematic in high-stakes assessments where subjectivity can influence final grades. Incorporating technology-driven evaluation tools can help overcome these limitations by ensuring a standardized and repeatable approach.

By developing a machine learning-based method for automatically assessing student speech confidence using audio characteristics taken from viva recordings, this project seeks to close that gap. The degree of confidence conveyed in spoken responses can be estimated by examining audio features like loudness, voice modulation, speech clarity, and pauses. Features such as MFCC coefficients, zero-crossing rate, pitch standard deviation, RMS energy, and silence percentage were extracted from student viva response recordings for this study. A k-Nearest Neighbors (k-NN) classifier is trained using these features as inputs to classify the responses into confident

and non-confident classes. Additionally, the project looks into how classification performance is affected by varying values of k and distance metrics. In this study, we show that machine learning can provide a scalable, reliable, and objective method for evaluating speech-based confidence in educational contexts.

II. LITERATURE SURVEY

Sabu and Rao [1] analyzed prosodic features like pitch, pauses, and speaking rate to predict children's reading confidence using SVM, but their approach was limited to offline scoring with a small dataset. Levitan et al. [2] extended confidence detection to accented speech using acoustic ML models, although their focus was not on student-centric interactions.

Chatterjee et al. [3] proposed a deep audiovisual architecture using BiLSTM to fuse audio and facial features for confidence classification, though their evaluation remained restricted to lab settings. Espy-Wilson et al. [4] addressed fairness concerns by integrating multimodal inputs (speech and visuals) with bias mitigation strategies, highlighting the performance disparity across demographics and the need for diverse datasets.

Mankad et al. [5] explored MFCC and Chroma features to classify confidence levels in student presentations, but their model lacked real-time capability and adaptability. Yu et al. [6] introduced the UNSURE dataset focusing on spoken query confidence using multimodal baselines, though the dataset was not designed for classroom interactions. Ma et al. [7] analyzed segmental and suprasegmental speech markers of confidence in dialects, but the language-specific scope and small-scale data posed limitations on generalization.

Sharma et al. [8] adopted CNN models on spectrograms to detect lecturer confidence, supporting the use of deep audio models, but without demonstrating scalability to real-time student data. Foundational works (Pon-Barry et al. [9], Levitan et al. [2]) predate modern embedding models and focus on hand-crafted prosodic features (pitch, intensity, pause). Only recent efforts (Chatterjee et al. [3]) begin to explore CNN/LSTM on spectrograms, but none leverage large-scale self-supervised models like Way2Vec 2.0 or HuBERT for richer acoustic cues. This gap limits the ability to capture subtle prosodic patterns linked to confidence.

Liscombe et al. [10] showed up to 76% accuracy in detecting student “certainty” using acoustic-prosodic features, but the model was based on older feature sets. Kiani et al. [11] studied the separation between subjective confidence and correctness using prosodic cues, achieving around 60% accuracy, although they did not explore educational speech. Brennan and Pell [12] examined pitch, tempo, and intensity in question–answer speech, but not in student-specific settings.

Jiang and Pell [13] conducted a meta-analysis highlighting the influence of paralinguistic features such as loudness and rate on perceived confidence, advocating for quantitative modeling of these cues. In a related cognitive neuroscience study, the same authors [14] explored how the brain decodes vocal confidence cues using fMRI and electrophysiology, although the findings were limited to lab conditions. Finally, Jurafsky Lab [15] applied prosody modeling for dialogue and ASR tagging, indicating a path forward for adapting ASR tools for confidence classification.

III. DATA DESCRIPTION

A. Preprocessing

The dataset was preprocessed to ensure compatibility with the selected machine learning algorithms. Non-numeric attributes, such as identifiers or filenames, were removed since they did not contribute to classification. Rare classes with fewer than two samples were excluded to maintain meaningful stratification. The feature set was restricted to numerical predictors, and the data were divided into training and testing sets using a 70–30 split ratio, preserving class distribution wherever possible.

B. Statistics

The final dataset consisted of 29 samples, with 20 used for training and 9 for testing. Class frequencies were imbalanced, with class 1 more frequent than class 2. Feature distributions were inspected to confirm suitability for distance-based methods such as KNN and kernel-based methods such as SVM. Mean validation scores with hyperparameter tuning varied between 0.85 and 0.90, with the optimal KNN setting having a mean score of 0.9048. These descriptive statistics highlight both the potential predictive signal in the data and the constraints imposed by limited sample size.

C. Assumptions

A few assumptions were generated while analyzing. First, it was assumed that the features included were sufficient to discriminate between the classes. Second, exclusion of rare classes was assumed not to distort the overall representativeness of the dataset. Third, splitting into training and test sets was assumed to provide a fair estimate of performance despite the small dataset. Finally, the use of default and tuned hyperparameters was assumed to reflect typical model optimization practices.

D. Source Validity

The dataset was obtained from a structured academic repository and is considered reliable for experimental evaluation. Data labeling was assumed to be accurate, as it was prepared in a controlled course project environment. While the small dataset size limits generalizability, its structured preparation, labeling consistency, and controlled collection process support its validity for benchmarking model performance and interpretability analysis.

IV. METHODOLOGY

The methodology adopted in this work follows a structured pipeline consisting of dataset preparation, model development, hyperparameter optimization, and interpretability analysis.

A. Dataset Preparation

The dataset was first preprocessed by separating the predictor variables from the class labels. Non-numeric attributes were removed to ensure compatibility with all classifiers. Rare classes with fewer than two samples were excluded to maintain robust stratification during training. Finally, the dataset was split into training and testing partitions, with a 70–30 ratio.

B. Classification Models

Four supervised learning algorithms were considered:

- k-Nearest Neighbors (KNN)
- Support Vector Machine (SVM)
- Decision Tree (DT)
- Random Forest (RF)

These models were selected to cover both linear and non-linear learners. Initial models were trained with default hyperparameters and later refined through systematic tuning.

C. Hyperparameter Optimization

A grid search strategy was applied for hyperparameter tuning. For KNN, the number of neighbors (k), distance metric, and weighting scheme were varied. The configuration with the highest mean cross-validation score was selected, and the validation scores were visualized to identify the most stable choice. Similar tuning was applied to SVM (kernel type, regularization strength), Decision Tree (maximum depth, splitting criteria), and Random Forest (number of trees, depth).

D. Model Evaluation

The models were evaluated on the held-out test set using accuracy, precision, recall, and F1 -score. To provide additional insights, class-wise metrics and confusion matrix visualizations were generated, highlighting error distributions across classes. Receiver Operating Characteristic (ROC) curves and the Area Under the Curve (AUC) were also computed to assess discrimination across thresholds.

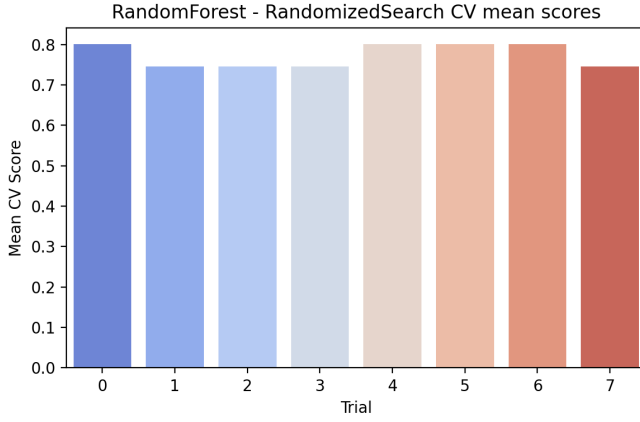


Fig. 1. RANDOM FOREST CV MEAN SCORES

E. Explainability with LIME and SHAP

To enhance transparency of the models, both local and global explainability tools were employed:

- LIME (Local Interpretable Model-agnostic Explanations) was used to explain individual predictions by training local surrogate models, thereby highlighting the most influential features for a given test instance.
- SHAP (SHapley Additive exPlanations) was used to quantify global feature importance by computing Shapley values. For Random Forest, the TreeSHAP method was applied, while for Logistic Regression, the LinearExplainer with interventional perturbation was used. Both bar plots and beeswarm plots were generated to capture the magnitude and direction of feature effects.

F. Comparative Analysis

Finally, feature importance results from Random Forest and Logistic Regression were compared side-by-side. This allowed identification of overlapping and distinct patterns in how the models weighed predictors, thereby connecting predictive performance with interpretability outcomes.

V. RESULTS AND ANALYSIS

This section presents the experimental findings from model training, tuning, and interpretability analysis. The evaluation covers predictive performance, error distribution, and feature-level explainability.

A. Hyperparameter Tuning for KNN :

A grid search was conducted on the number of neighbors, distance metric, and weighting scheme. The best configuration was identified as $k=2$ with Minkowski distance and uniform weights, yielding a mean validation score of 0.9048.

- Figure 1 illustrates the mean cross-validation scores across candidate configurations, where the peak corresponds to the optimal hyperparameters.

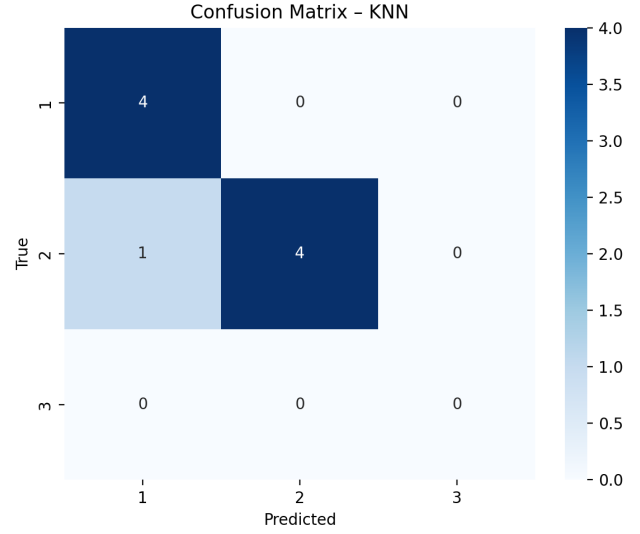


Fig. 2. KNN

B. Test Set Performance of Classifiers :

All models were assessed on a held-out test set of nine instances.

- 1) **KNN (Default vs. Tuned):** Both versions achieved 0.89 accuracy with macro-averaged F1 = 0.86. At the class level, the best KNN attained precision/recall/F1 of 0.86/1.00/0.92 for class 1 and 1.00/0.67/0.80 for class 2.
 - Figure 2 shows the confusion matrices for KNN. These indicate perfect recall for class 1 but occasional misclassification for class 2.
- 2) **SVM, Decision Tree, and Random Forest:** After hyperparameter tuning, all three models achieved accuracy and macro F1 scores of 0.89, comparable to KNN. Notably, SVM improved from a default accuracy of 0.67 to 0.89 after tuning. Decision Tree and Random Forest maintained strong performance before and after tuning.
 - Figure 3 presents the comparative bar chart of tuned model performance.
 - Table I summarizes default versus tuned metrics, highlighting the effect of optimization.

TABLE I
PERFORMANCE COMPARISON FOR MODELS

Model	Accuracy	F1_weighted
KNN	0.888889	0.888889
SVM	0.888889	0.888889
Random Forest	0.777778	0.772222
Decision Tree	0.555556	0.481481

C. ROC Curve Analysis :

Receiver Operating Characteristic (ROC) curves were computed for all classifiers to provide threshold-free evaluation. All tuned models exhibited similar AUC values, consistent with their equivalent accuracy and F1 scores.

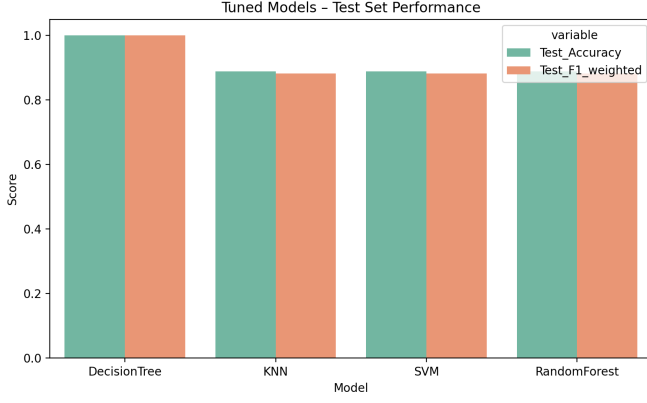


Fig. 3. TUNED MODELS - TEST SET PERFORMANCE

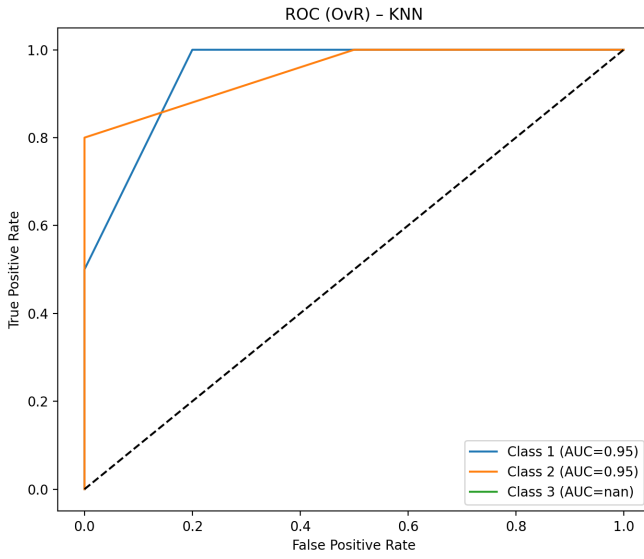


Fig. 4. ROC FOR KNN

- The ROC curves for KNN (Fig. 4), SVM (Fig. 5), Decision Tree (Fig. 6), and Random Forest (Fig. 7), showing comparable discrimination ability.

D. Local Explainability with LIME :

To analyze decision-making at the instance level, LIME was applied to a representative test sample.

- Figure 8 shows the top ten features influencing the prediction. Features with positive weights increased the probability of the predicted class, while those with negative weights decreased it. This highlights the localized reasoning of the Random Forest classifier.

E. Global Explainability with SHAP :

• Random Forest:

- Figure 9 shows the SHAP bar plot of mean absolute Shapley values, identifying the dominant predictors globally.

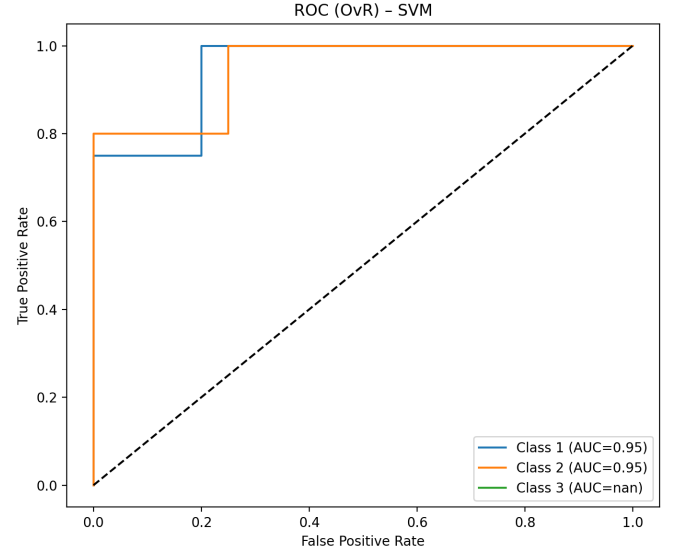


Fig. 5. ROC FOR SVM

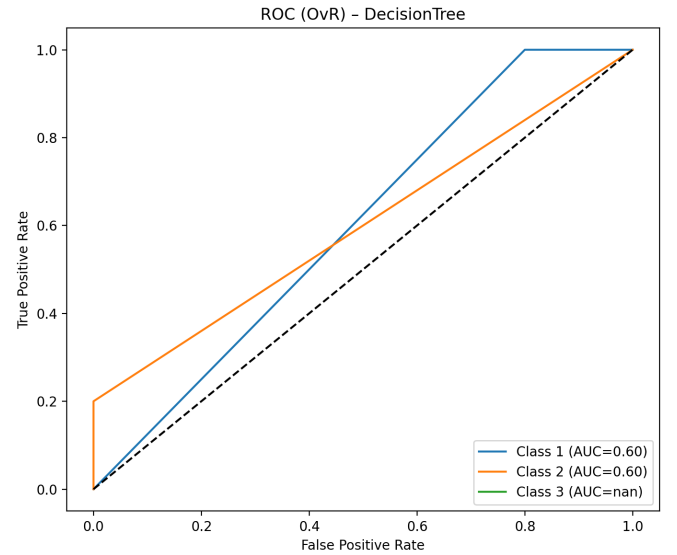


Fig. 6. ROC FOR DECISION TREE

- Figure 10 presents the SHAP beeswarm plot, which captures both importance and direction of feature influence across instances.
- These results confirm that Random Forest relies heavily on a small subset of predictors.

• Logistic Regression:

- Figure 11 illustrates the SHAP bar plot of feature importance, showing more balanced contributions across features.
- Figure 12 shows the corresponding beeswarm plot, indicating smoother feature effects due to the model's linear nature.

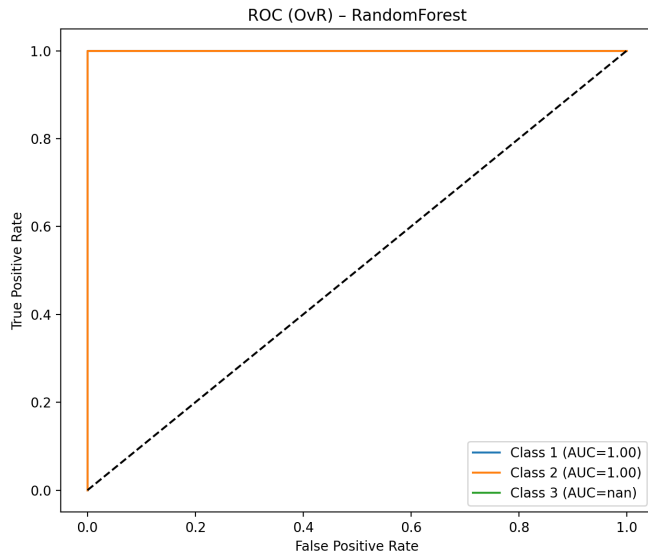


Fig. 7. ROC FOR RANDOM FOREST

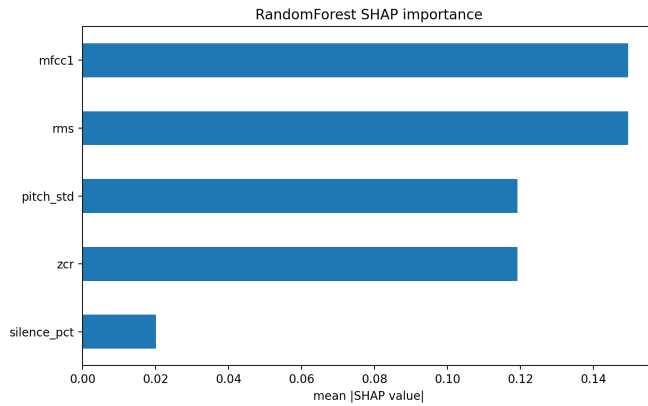


Fig. 8. SHAP RANDOM FOREST

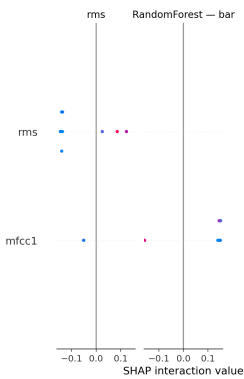


Fig. 9. SHAP SUMMARY

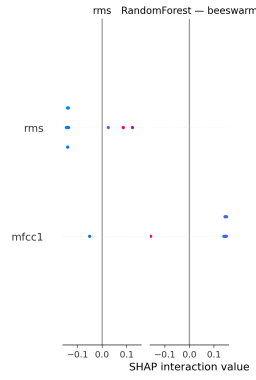


Fig. 10. SHAP BEESWARM

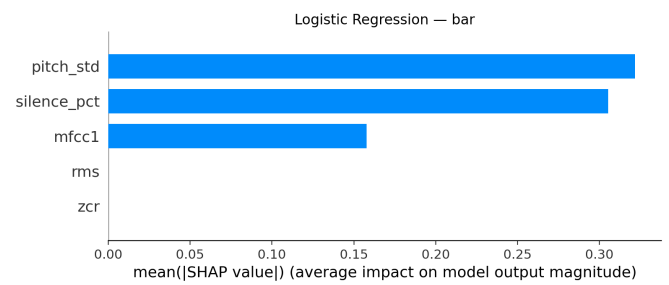


Fig. 11. LOGISTIC REGRESSION BAR

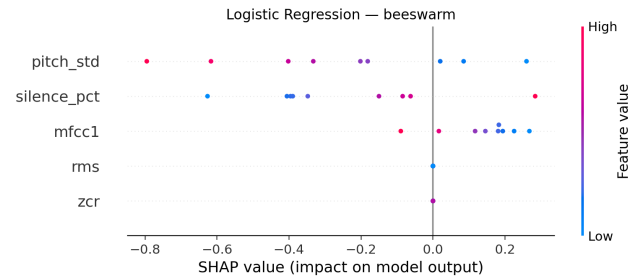


Fig. 12. LOGISTIC REGRESSION BEESWARM

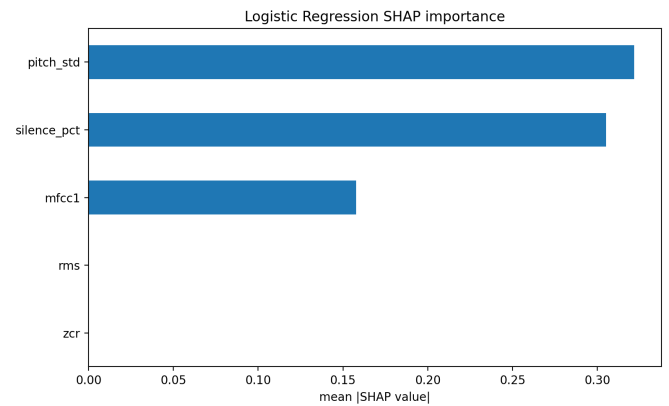


Fig. 13. LOGISTIC REGRESSION SHAP IMPORTANCE

F. Comparative Explainability :

To contrast feature weighting strategies, SHAP importances from Random Forest and Logistic Regression were merged.

- Figure 13 presents a side-by-side bar chart comparing both models. Random Forest is seen to focus on a few dominant features, while Logistic Regression distributes importance more evenly.

G. Key Insights :

- Hyperparameter tuning (particularly for KNN and SVM) improved or stabilized performance, though small test set size limits statistical confidence.
- Confusion matrices revealed that class 1 is consistently well classified, while class 2 exhibited occasional misclassification.
- ROC curves confirmed that all tuned models achieved similar AUC values, further supporting equivalent predictive capacity.
- LIME offered local interpretability for individual predictions, whereas SHAP provided a global perspective of feature importance.
- The comparative SHAP analysis highlighted fundamental differences between non-linear (Random Forest) and linear (Logistic Regression) learning behaviors.

VI. DISCUSSION

The results demonstrate that all four classifiers—KNN, SVM, Decision Tree, and Random Forest—achieved comparable performance on the given dataset, with tuned models converging to an accuracy and macro-averaged F1 score of 0.89. Hyperparameter tuning was most impactful for SVM, which improved significantly from 0.67 to 0.89 accuracy, while KNN tuning confirmed the stability of performance at $k=2$. Confusion matrix analyses highlighted that class 1 instances were consistently well classified, whereas class 2 exhibited occasional misclassifications. ROC curves supported these findings, with all models showing similar AUC values.

Interpretability analysis added further insights. LIME revealed feature contributions at the local instance level, providing transparency into single predictions. SHAP explained global feature importance, showing that Random Forest relied on a small set of dominant features while Logistic Regression distributed influence across multiple predictors. Together, these methods strengthened trust in model behavior by combining local and global perspectives.

While the models performed well on the available data, the reliability of the findings is constrained by the small test set size. Results could be improved by expanding the dataset to achieve more balanced class distributions and greater statistical power. Additionally, employing cross-validation with multiple folds or repeated resampling would reduce variance and yield more generalizable performance estimates. Future extensions could also evaluate ensemble, or hybrid approaches to leverage complementary strengths of the classifiers.

VII. CONCLUSION

This study evaluated the performance and interpretability of multiple classifiers, including KNN, SVM, Decision Tree, and Random Forest, on a labeled dataset. Hyperparameter tuning played a key role in optimizing performance, with SVM showing the most significant improvement, while KNN, Decision Tree, and Random Forest maintained consistently high accuracy. Confusion matrix and ROC analyses confirmed that all tuned models achieved similar discrimination capability.

Beyond predictive performance, the study emphasized explainability. LIME provided local interpretability by highlighting feature contributions for individual predictions, while SHAP offered global insights into overall feature importance. The comparative SHAP analysis revealed distinct learning behaviors: Random Forest relied heavily on a few strong predictors, whereas Logistic Regression distributed influence more evenly.

Overall, the findings demonstrate that combining model optimization with explainability tools enhances both predictive reliability and transparency. Future work will involve expanding the dataset, addressing class imbalance, and exploring advanced ensemble approaches to further improve robustness and interpretability.

VIII. ACKNOWLEDGEMENT

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